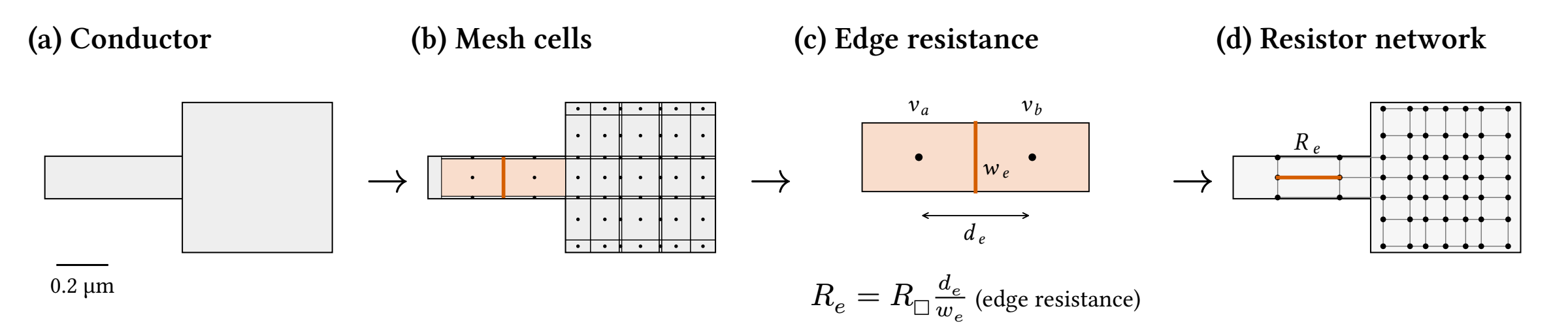
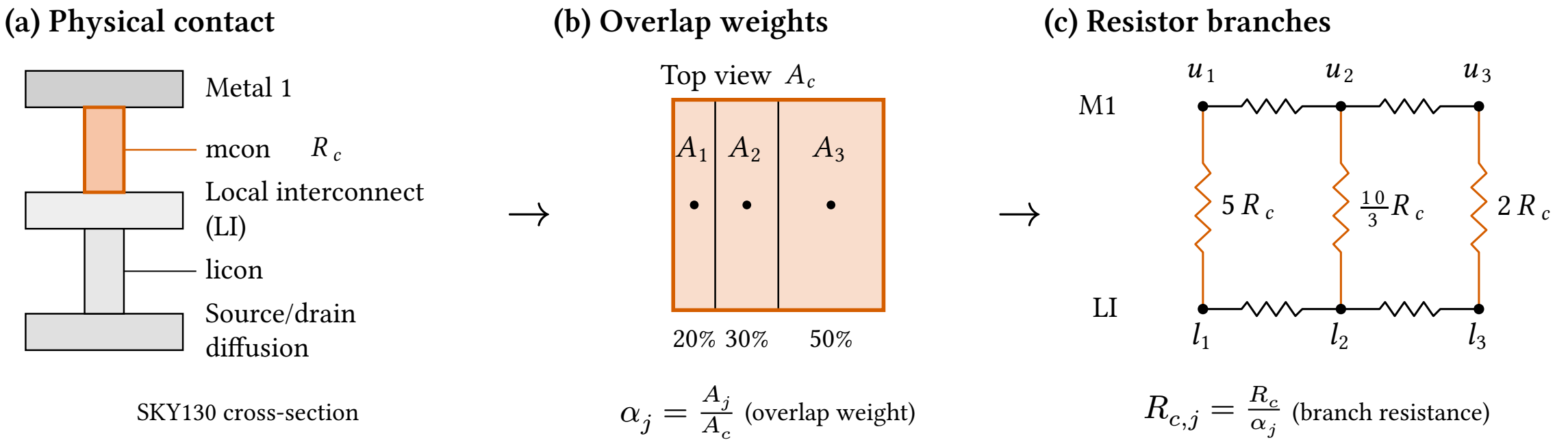
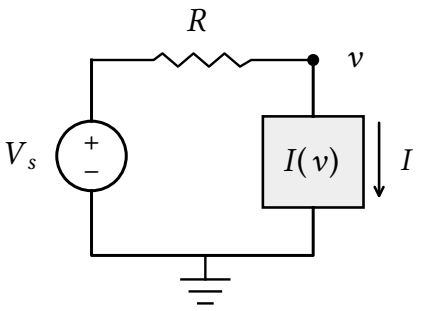


$$R_e = R_\square \frac{d_e}{w_e} \text{ (edge resistance)}$$


(a) Physical contact
Metal 1
mcon
R_c
Local interconnect (LI)
licon
Source/drain diffusion
SKY130 cross-section
(b) Overlap weights
Top view A_c
A_1
A_2
A_3
20%
30%
50%
$\alpha_j = \frac{A_j}{A_c}$ (overlap weight)
(c) Resistor branches
u_1
u_2
u_3
M1
LI
$5\,R_c$
$\frac{10}{3}R_c$
$2\,R_c$
l_1
l_2
l_3
$R_{c,j} = \frac{R_c}{\alpha_j}$ (branch resistance)

(a) Fixed circuit



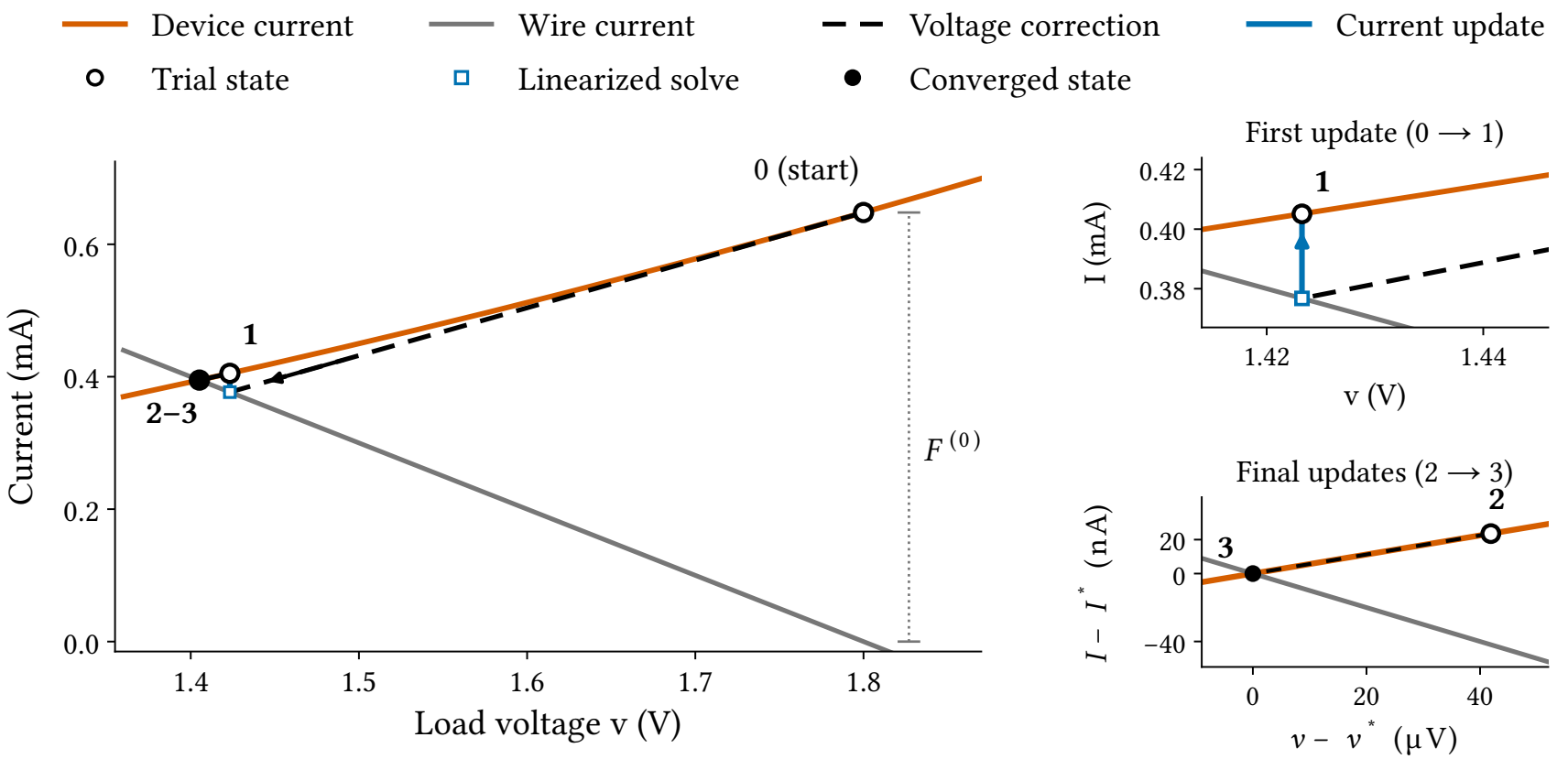
$V_s = 1.8\,\mathrm{V}, \quad R = 1\,\mathrm{k\Omega}$

$I(v) = 0.2\,\mathrm{mA}\left(\frac{v}{1\,\mathrm{V}}\right)^2$

v^* (converged voltage)
1.405125 V

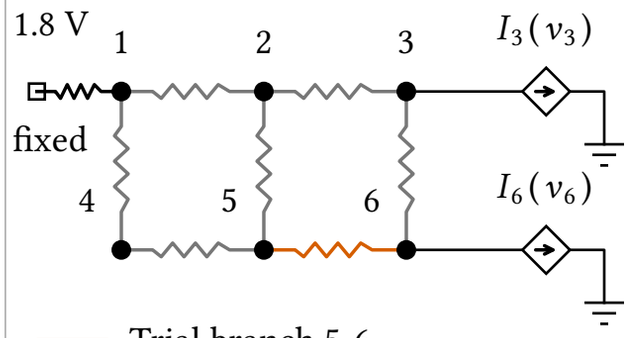
I^* (converged current)
0.394875 mA

(b) Voltage–current iteration



$I_{\text{wire}} = \frac{V_s - v}{R}$ (wire current) $\qquad I_{\text{device}} = I(v)$ (device current) $\qquad F = I_{\text{device}} - I_{\text{wire}}$ (current mismatch)

(a) Baseline circuit



$$\Delta g_{56} = +0.1\,\text{mS} \quad (+1\%)$$

$$\Delta I = \left(v_5^{\text{base}} - v_6^{\text{base}}\right)\Delta g_{56}$$

Equivalent current disturbance

$$\Delta I = 6.877\,\mu\text{A}$$

Known at baseline

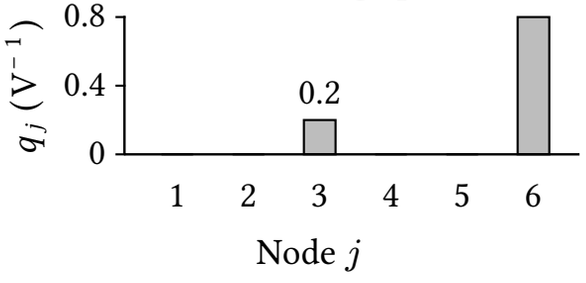
$$K = G + \frac{\partial \boldsymbol{i}_{\text{load}}}{\partial \boldsymbol{v}}$$

Wire matrix + load slopes

$$m = \frac{0.2v_3 + 0.8v_6}{1\,\text{V}}$$

$$q_j = \frac{\partial m}{\partial v_j}$$

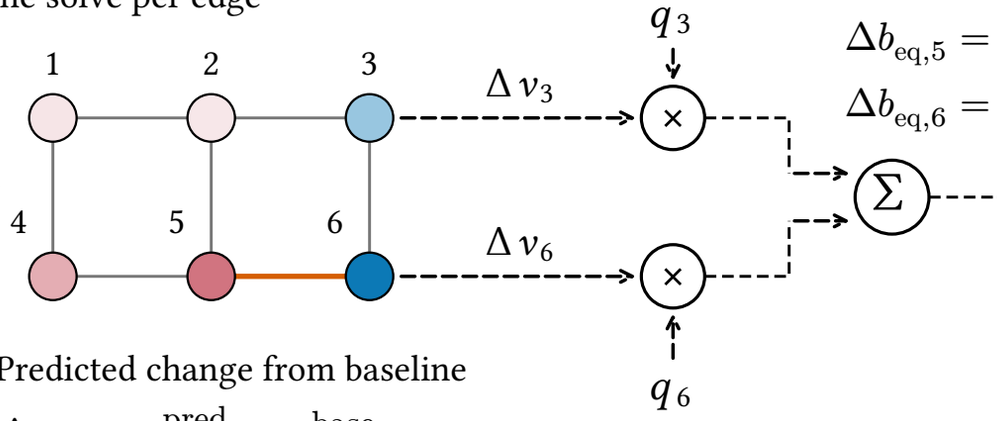
Metric change per volt



(b) Direct $K\Delta v = \Delta b_{\text{eq}}$

Calculation flow

One solve per edge



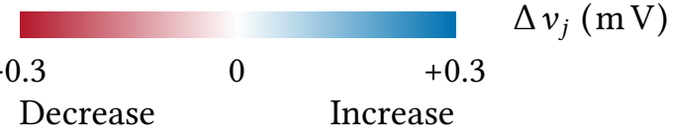
$$\Delta b_{\text{eq},5} = -\Delta I$$

$$\Delta b_{\text{eq},6} = +\Delta I$$

Others zero

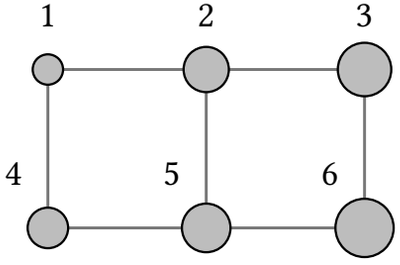
Predicted change from baseline

$$\Delta v_j = v_j^{\text{pred}} - v_j^{\text{base}}$$



(c) Adjoint $K^T\boldsymbol{\lambda} = \boldsymbol{q}$

One solve per metric

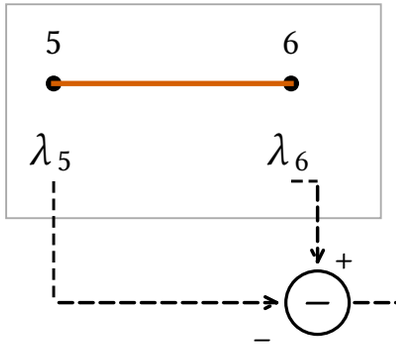


Metric change per injected ampere

Circle area $\propto \lambda_j$ (A^{-1})



Read branch 5-6



Reuse for any branch

Comparison

First-order Δm

Direct

$$+2.53 \times 10^{-4}$$

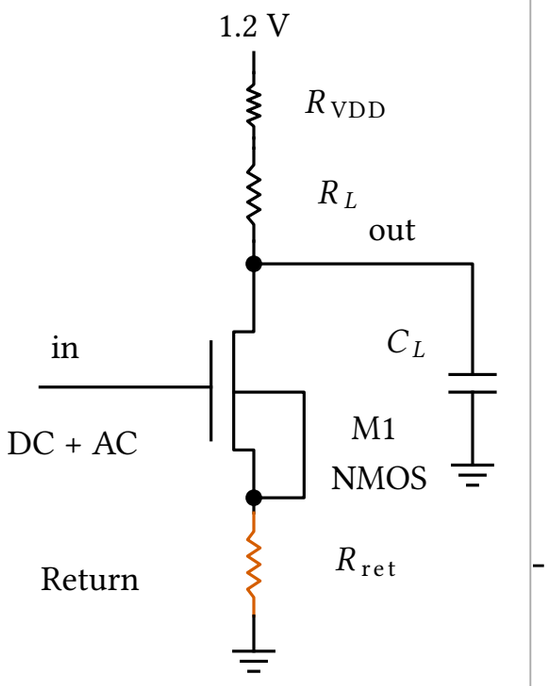
=

Adjoint

$$+2.53 \times 10^{-4}$$

Dimensionless

(a) Common-source



$w : 0.17 \rightarrow 0.49\,\mu m$

(b) Two effects

Wire effect

Device parameters held fixed

ΔR_i → $g_m,\ g_{ds},\ C$ → Fixed-bias AC response → AC metric

Bias effect

DC terminal voltages change

$K,\ \Delta \mathbf{v}_{DC}$ → $\Delta g_m,\ \Delta g_{ds},\ \Delta C$ → Device model changes → AC metric

$\Delta A = 4\,\mu m^2$

--> Effect flow

(c) Contributions

First-order metric changes

	Wire	Bias	Total
ΔG (dB)	+0.9626	+0.0405	**+1.0031**
Δf_u (MHz)	+4.186	+0.5604	**+4.7463**

g_m (transconductance)

g_{ds} (output conductance)

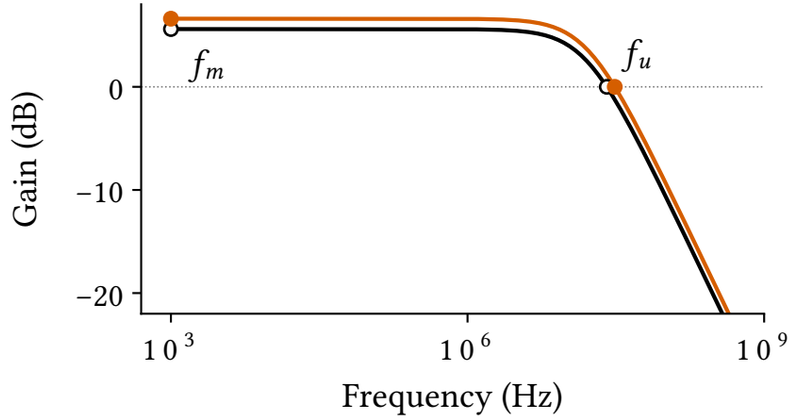
C (device capacitance)

$H(f) = \frac{\tilde{V}_{out}}{\tilde{V}_{in}}$ AC voltage ratio

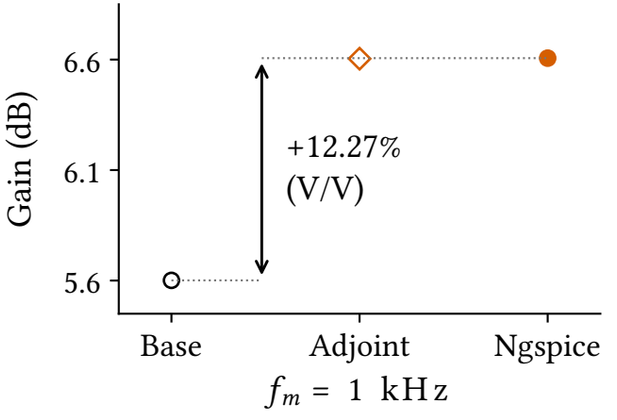
(d) Native verification

Baseline ◇ First order Ngspice widened

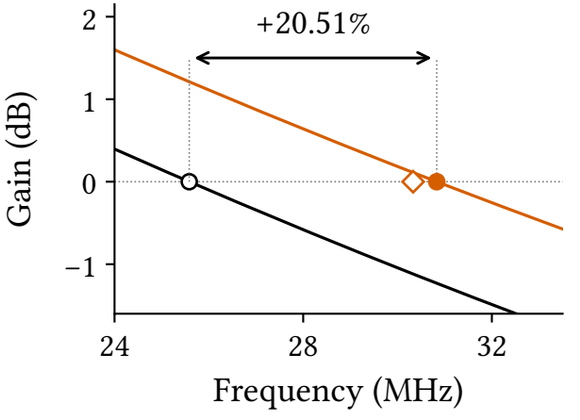
$G(f) = 20\log_{10}|H(f)|$



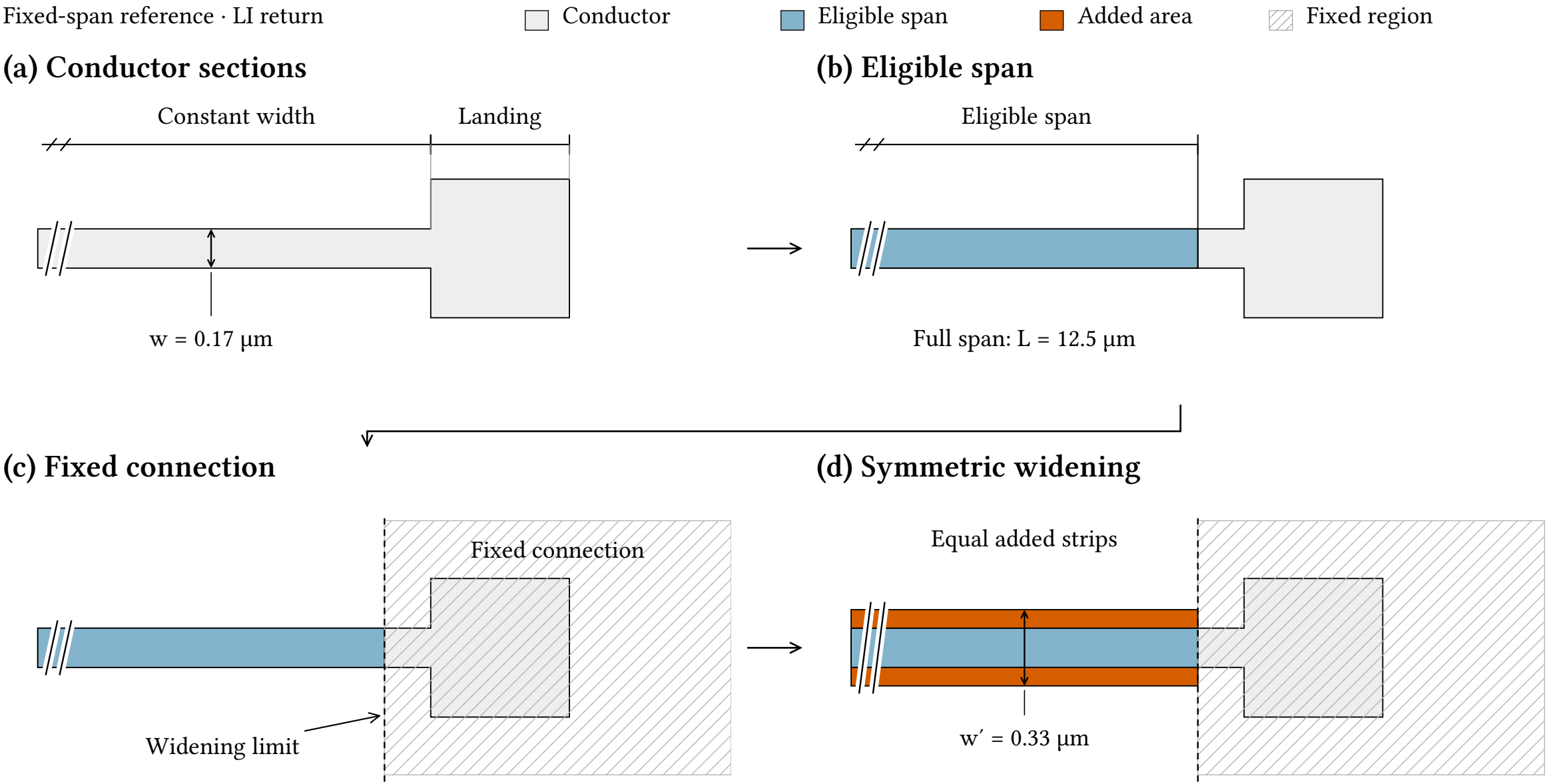
Gain at f_m



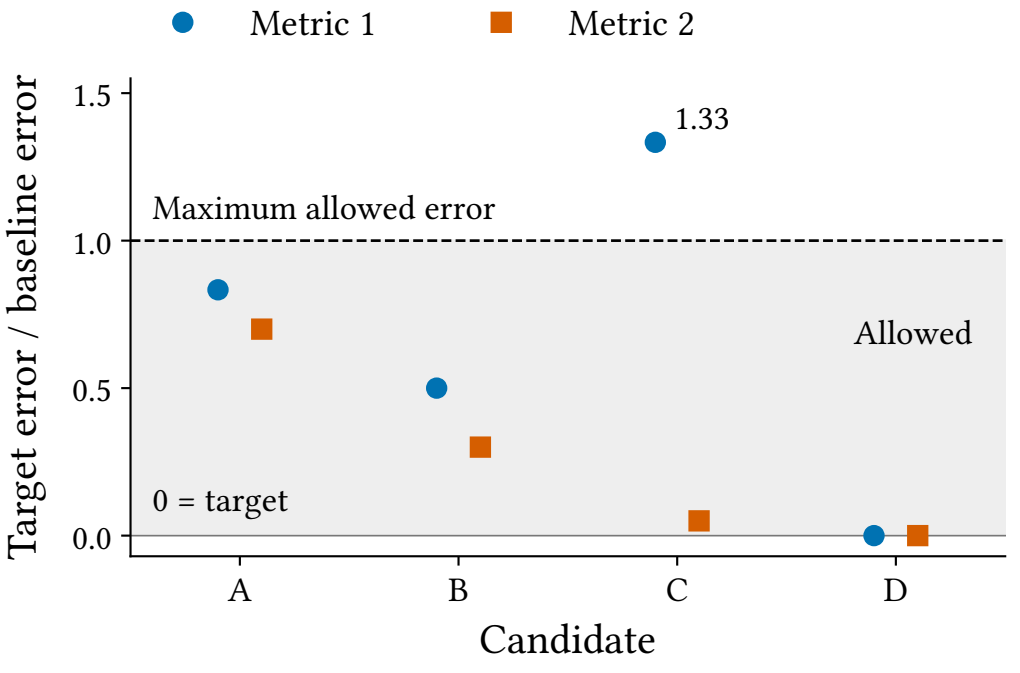
Unity: $G(f_u) = 0$



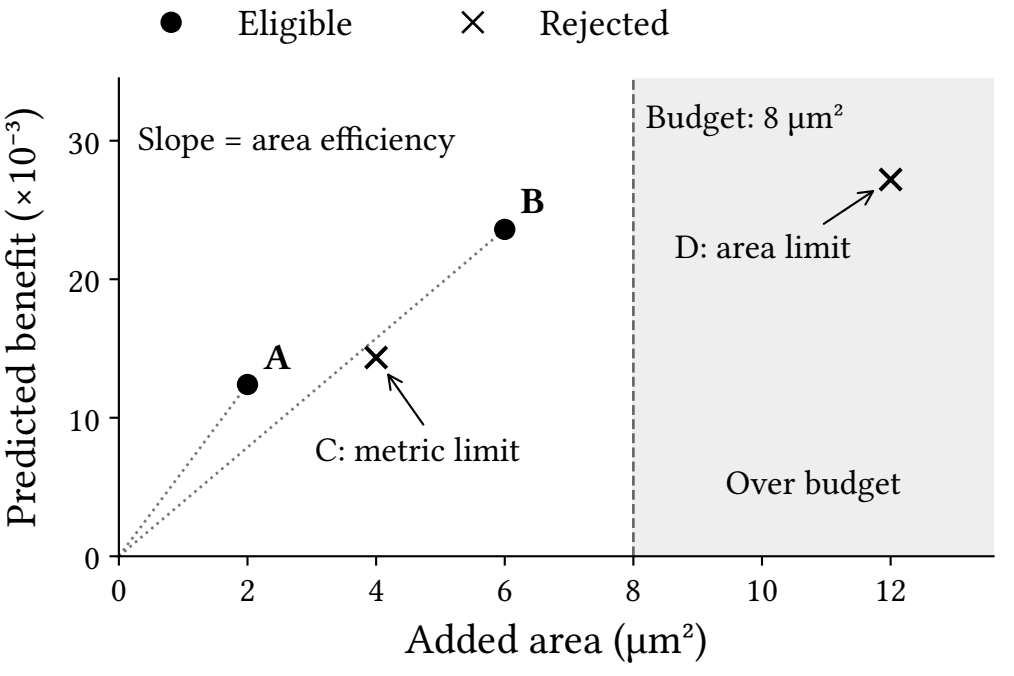
Ngspice DC + AC → AC adjoint → DC bias adjoint → Per-trace sensitivities


Fixed-span reference · LI return
Conductor
Eligible span
Added area
Fixed region
(a) Conductor sections
Constant width
Landing
w = 0.17 µm
(b) Eligible span
Eligible span
Full span: L = 12.5 µm
(c) Fixed connection
Fixed connection
Widening limit
(d) Symmetric widening
Equal added strips
w′ = 0.33 µm
0.5 µm
$\Delta A = L(w' - w) = (12.5\,\mu\text{m})(0.16\,\mu\text{m}) = 2\,\mu\text{m}^2$ (added area)

(a) Metric limits



(b) Area and benefit



(c) Total-benefit ranking

Benefit × 10^{-3} · Per area × 10^{-3} µm^{-2}

1. B	23.60	3.93	Preferred	**2. A**	12.40	6.20	More efficient

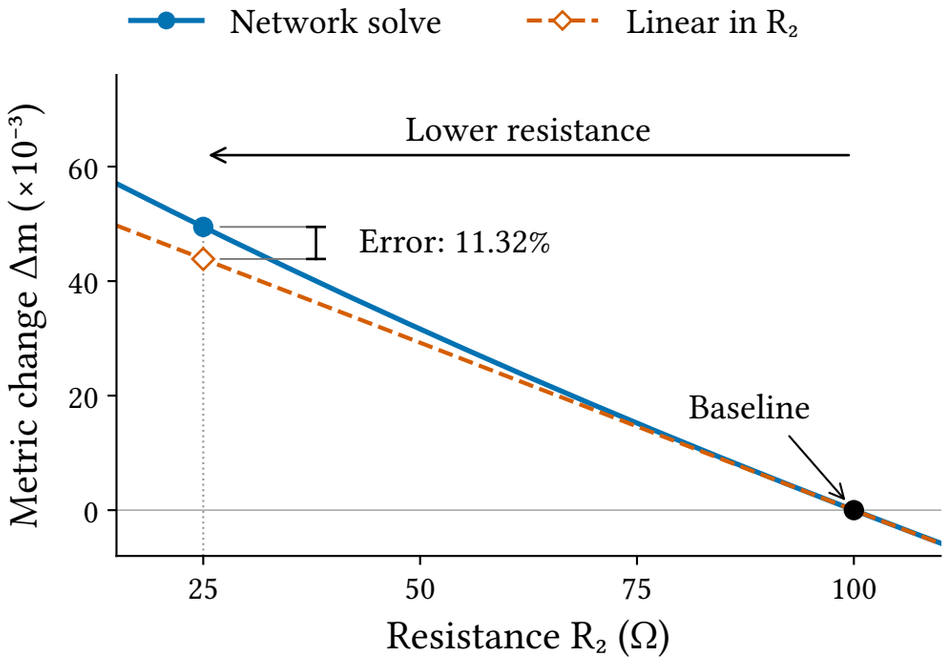
$$J = \tfrac{1}{2}\sum_k (m_k - 1)^2$$
(combined error)

$$\hat{B}_a = J_0 - \hat{J}_a$$
(predicted benefit)

$$\hat{\eta}_a = \frac{\hat{B}_a}{\Delta A_a}$$
(area efficiency)

Illustrative resistor networks

(a) Error in predicted change

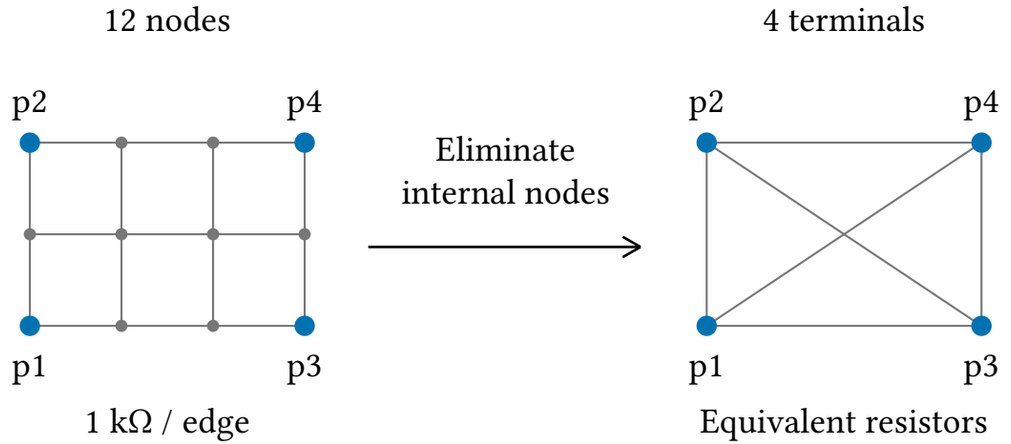


$\Delta m_{\text{pred}} = 0.043868$ (linear in R_2)

$\Delta m_{\text{solved}} = 0.049468$ (network solve)

Error relative to solved change

(b) Terminal reduction



Same terminal response

Port	V (V)	Full (µA)	Reduced (µA)
p1	1.0	+334.0	+334.0
p2	0.8	+34.0	+34.0
p3	0.6	-34.0	-34.0
p4	0.4	-334.0	-334.0

Current into network Mismatch < 1 pA

